Infant State Recognition System: Multi-Class Cry Classification via Neuroscience-Informed Acoustic Feature Engineering and Classical Machine Learning

Phase 1 Report — Advanced Machine Learning Foundation
with Data Leakage Analysis and Correction

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Abstract

Infant cry is the primary pre-linguistic communication channel between neonates and caregivers. While experienced clinicians can distinguish hunger cries from pain cries with moderate reliability, automated systems promise consistent, round-the-clock monitoring in both clinical and consumer IoT settings. This report presents Phase 1 of a three-phase hybrid ML/DL system for five-class infant cry classification (*hunger*, *belly pain*, *burping*, *discomfort*, *tiredness*) using the Donate-a-Cry corpus. We introduce a neuroscience-informed 411-dimensional feature vector that fuses Mel-Frequency Cepstral Coefficients (MFCCs), Constant-Q Cepstral Coefficients (CQCCs), fundamental frequency (F0) contour statistics, spectral contrast, and chroma features. Six classical models are evaluated: Bayesian Gaussian Mixture Model (GMM), SMOTE-enhanced Support Vector Machine (SVM), Hidden Markov Model (HMM) with left-right topology, Random Forest (RF), XGBoost, and a stacking ensemble. We perform rigorous statistical validation of distributional assumptions (ADF stationarity, Shapiro–Wilk Gaussianity), systematic hyperparameter tuning with macro-F1 optimisation, and detailed failure analysis. Critically, we identify and correct a **data leakage** issue where augmented audio variants leaked across the train/test split, inflating initial results (XGBoost: 96.5% accuracy). After correction via a split-before-augment protocol, the Ensemble achieves 83.7% accuracy and SVM achieves the best macro-F1 of 0.270 on the leak-free test set of 92 original recordings. This honest evaluation reveals the true challenge posed by extreme class imbalance (47:1 ratio) and establishes a rigorous baseline for Phase 2 (deep learning) and Phase 3 (hybrid fusion with edge deployment).

1 Introduction

Crying is the earliest and most instinctive form of vocal communication for human infants. The periaqueductal gray (PAG) region of the midbrain generates acoustically distinct cry patterns depending on the underlying physiological state [2]. Dunstan Baby Language theory identifies five pre-cry reflexes—*Neh* (hunger), *Owh* (tiredness), *Heh* (discomfort), *Eairh* (lower gas/belly pain), and *Eh* (burping)—each driven by distinct neuromuscular pathways [1].

From a signal processing perspective, infant cries exhibit three distinct phonation modes: (i) *modal voice* (fundamental frequency $F_0 \approx 250$–$600\,\mathrm{Hz}$), characteristic of hunger and mild

discomfort; (ii) *hyperphonation* ($F_0 > 1000\,\text{Hz}$), associated with acute pain; and (iii) *dysphonation* (turbulent, aperiodic), indicating severe distress [3]. These acoustic signatures motivate our choice of features that capture both spectral envelope (MFCC, CQCC) and prosodic dynamics (F0 contour, voiced fraction).

Contributions of this phase.

1. A **411-dimensional neuroscience-informed feature vector** combining MFCCs (240 dim), CQCCs (120 dim), F0 contour statistics (7 dim), spectral contrast (14 dim), chroma (24 dim), and spectral descriptors (6 dim).
2. **Six classical models** with principled imbalance handling: Bayesian GMM, SMOTE-SVM with grid search, left-right HMM, Random Forest, XGBoost, and a stacking meta-learner.
3. **Statistical validation** of stationarity (ADF) and Gaussianity (Shapiro–Wilk) assumptions.
4. **Failure analysis** identifying specific instances and class confusions where models fail, with mathematical explanations rooted in acoustic similarity.
5. A **17-technique data augmentation** pipeline that addresses extreme class imbalance (47:1 ratio between hunger and burping).

2 Related Work

2.1 The Neuroscience of Infant Cry

The generation of infant cry is mediated by the PAG, whose dorsolateral and lateral columns produce acoustically distinct vocalisation patterns [2]. Vagal tone—measured via heart rate variability—modulates cry acoustics: low vagal tone correlates with higher F_0 and more dysphonation [4]. This neurophysiological grounding suggests that F_0 dynamics and spectral regularity are not arbitrary features but direct acoustic correlates of the infant's autonomic state.

2.2 Acoustic Features for Cry Analysis

MFCCs. Mel-frequency cepstral coefficients remain the dominant feature for audio classification [7]. The mel-scale frequency warping approximates the human auditory system's nonlinear frequency perception, making MFCCs effective at capturing the spectral envelope shaped by the infant's vocal tract.

CQCCs. Todisco *et al.* [8] introduced Constant-Q Cepstral Coefficients, which apply a Constant-Q Transform (CQT) instead of the linear-frequency FFT. The CQT provides logarithmically-spaced frequency bins with variable time-frequency resolution—higher frequency resolution at low frequencies where infant cry harmonics are concentrated, and higher temporal resolution at high frequencies where transient events matter. Notably, Maghfira *et al.* [9] achieved 99.82% accuracy on the Baby Chillanto dataset using GMM with CQCC features, establishing CQCCs as superior to MFCCs for infant cry classification.

Prosodic features. Fundamental frequency (F_0) contour analysis captures the pitch dynamics that distinguish pain cries (rising F_0, hyperphonation) from hunger cries (stable, modal F_0) [3]. We extract seven statistics from the F_0 contour: mean, standard deviation, maximum, minimum, range, voiced fraction, and slope.

2.3 Classical ML for Audio Classification

Gaussian Mixture Models. GMMs model each class as a mixture of multivariate Gaussians, with classification via maximum likelihood [10]. We upgrade to Bayesian GMMs with a Dirichlet

process prior that automatically prunes unnecessary components, avoiding the BIC-based model selection step.

Support Vector Machines. SVMs with RBF kernels are effective for moderate-dimensional audio features [11]. However, severe class imbalance (47:1) causes SVMs to collapse to majority-class prediction—a phenomenon we address with SMOTE oversampling and One-vs-One decision functions.

Random Forests and Gradient Boosting. Salehian *et al.* achieved 96.39% accuracy using MFCC features with Random Forest [12], demonstrating that ensemble tree methods effectively capture nonlinear feature interactions in cry classification. XGBoost further regularises via column subsampling and shrinkage.

Hidden Markov Models. HMMs extend GMMs by modelling the temporal sequence of MFCC frames as a Markov chain [13]. We adopt a left-right (Bakis) topology that enforces temporal monotonicity—the cry signal progresses through states without revisiting earlier ones.

2.4 Gap Analysis

Existing systems suffer from three critical limitations:

1. **Feature poverty**: Most systems use MFCCs alone, ignoring the variable-resolution advantage of CQCCs and the diagnostic value of F_0 dynamics.
2. **Imbalance blindness**: Published accuracies (e.g., 83.7%) mask catastrophic per-class failures (macro-F1 = 0.25) when datasets are imbalanced.
3. **No deployment pathway**: Models are evaluated in isolation without considering edge constraints (ESP32, browser-based inference).

This project addresses all three gaps: CQCC-enriched features, SMOTE-based imbalance correction evaluated via macro-F1, and an architecture designed for Phase 3 edge deployment.

3 Methodology

3.1 Dataset

The Donate-a-Cry corpus [15] contains short audio recordings of infant cries organised into five classes aligned with Dunstan Baby Language categories. Each clip is loaded at 8 kHz (matching the Nyquist frequency for the relevant cry bandwidth < 4 kHz) and normalised to 7 seconds via zero-padding or truncation.

Table 1: Dataset composition after augmentation.

Class	Original	After Aug.	Augmentation Factor
hunger	382	382	1.0× (dominant, not augmented)
belly_pain	18	288	16.0×
burping	8	144	18.0×
discomfort	22	300	13.6×
tiredness	27	300	11.1×
Total	**457**	**1414**	—

Imbalance analysis. The original dataset exhibits extreme class imbalance: the hunger class contains 382 clips while burping has only 8—a 47.75:1 ratio. Naive training on this distribution causes discriminative models (SVM) to achieve high accuracy (∼83%) by predicting the majority class for nearly all inputs, while macro-F1 collapses to ∼0.25.

3.2 Data Augmentation

We apply 17 augmentation techniques to minority classes, targeting 300 samples per class:

1. **Pitch shift**: $\pm 0.5, \pm 1, \pm 2$ semitones (6 variants)
2. **Time stretch**: 0.85×, 0.9×, 1.1×, 1.15× (4 variants)
3. **White noise injection**: SNR $\approx$ 25–30 dB
4. **Temporal shift**: ± 0.5 s with zero-padding
5. **Random gain**: ± 6 dB volume variation
6. **Bandpass filter**: Simulated microphone variation (200–3800 Hz)
7. **Reverb simulation**: Synthetic room impulse response ($RT_{60} \in [0.1, 0.4]$ s)
8. **Combination**: Pitch shift + noise (for non-trivial composites)

The dominant class (hunger) is *never* augmented to avoid worsening the imbalance.

Split-before-augment protocol. A critical methodological correction was applied after initial experiments revealed data leakage. In the initial pipeline, augmented files were written to disk alongside originals, and a subsequent random train/test split scattered augmented variants of the same original recording across both sets. This allowed models to be tested on near-copies of their training data, producing artificially inflated metrics (e.g., XGBoost: 96.5% accuracy). The corrected pipeline enforces:

1. Load only original recordings (457 files, excluding `_aug_` suffixed files).
2. Split originals into train (365) and test (92) with 80/20 stratification.
3. Augment *only* the training set in memory (no disk writes).
4. Extract features separately for train and test.

This ensures the test set contains *exclusively* original, never-augmented recordings, producing an honest evaluation.

3.3 Feature Extraction

Our 411-dimensional feature vector comprises six blocks, each motivated by a distinct acoustic property of infant cries:

3.3.1 Block 1: MFCC Statistics (240 dimensions)

We extract 40 MFCCs per frame using a 512-sample FFT window ($\approx$ 64 ms at 8 kHz) with 160-sample hop ($\approx$ 20 ms) and 64 mel bands. For each of the 40 coefficients, we compute six summary statistics (mean, std, min, max, skewness, kurtosis) over all time frames. The first-order (Δ) and second-order ($\Delta\Delta$) temporal derivatives are similarly summarised:

$$\mathbf{f}_{\text{MFCC}} = \big[\text{stats}(\mathbf{C}),\ \text{stats}(\Delta\mathbf{C}),\ \text{stats}(\Delta^2\mathbf{C})\big] \in \mathbb{R}^{240} \tag{1}$$

where $\text{stats}(\cdot) = [\mu, \sigma, \min, \max, \text{skew}, \text{kurt}]$.

3.3.2 Block 2: CQCC Features (120 dimensions)

The Constant-Q Transform provides geometrically-spaced frequency bins:

$$f_k = f_{\min} \cdot 2^{k/B}, \quad k = 0, 1, \ldots, K-1 \tag{2}$$

where $B = 12$ bins per octave (semitone resolution) and $K = 84$ bins spanning 7 octaves. The CQT magnitude spectrum is converted to log-power and projected onto a cepstral basis via the Type-II DCT:

$$\mathrm{CQCC}[n] = \sum_{k=0}^{K-1} \log |X_{\mathrm{CQ}}[k]|^2 \cdot \cos\left[\frac{\pi n(2k+1)}{2K}\right], \quad n = 0, \ldots, N_c - 1 \tag{3}$$

We retain $N_c = 20$ cepstral coefficients plus their Δ and $\Delta\Delta$ derivatives, summarised by mean and standard deviation: $20 \times 2 \times 3 = 120$ dimensions.

Why CQCCs outperform MFCCs for infant cry. Infant cry harmonics are concentrated in the 250–2000 Hz range. The CQT's logarithmic frequency resolution provides 3–4× finer frequency discrimination in this band compared to the linear FFT used by MFCCs, enabling better separation of closely-spaced harmonic overtones that distinguish hunger (modal $F_0 \approx 400$ Hz) from pain (hyperphonation $F_0 > 1000$ Hz).

3.3.3 Block 3: F0 Contour Statistics (7 dimensions)

Fundamental frequency is extracted via the probabilistic YIN (pYIN) algorithm [14]:

$$\mathbf{f}_{F_0} = [F_{0,\mu}, F_{0,\sigma}, F_{0,\max}, F_{0,\min}, F_{0,\mathrm{range}}, \rho_{\mathrm{voiced}}, \beta_{F_0}] \in \mathbb{R}^7 \tag{4}$$

where ρ_{voiced} is the fraction of voiced frames and β_{F_0} is the linear slope of the F_0 trajectory (rising slope $\Rightarrow$ escalating distress).

3.3.4 Block 4: Spectral Contrast (14 dimensions)

Spectral contrast measures the decibel difference between peaks and valleys in 7 frequency sub-bands (6 octave bands + 1 total):

$$\mathrm{SC}_b = \mathrm{Peak}_b - \mathrm{Valley}_b, \quad b = 1, \ldots, 7 \tag{5}$$

Summarised by mean and std: $7 \times 2 = 14$ dimensions. High contrast indicates harmonic richness (modal phonation); low contrast indicates noise-dominated spectra (dysphonation).

3.3.5 Block 5: Chroma Features (24 dimensions)

Chroma vectors map the spectrum onto 12 pitch classes, capturing harmonic structure independent of octave. Mean and std of 12 chroma bins yield 24 dimensions.

3.3.6 Block 6: Spectral Descriptors (6 dimensions)

Zero-crossing rate, spectral centroid, bandwidth, rolloff, and RMS energy, each summarised by mean only: 6 dimensions.

Table 2: Feature vector composition (411 dimensions total).

Block	**Description**	**Dim**
MFCC stats	40 MFCCs × 6 stats × 1 (static only)	240
CQCC stats	20 CQCCs × 2 stats × 3 (static+Δ+Δ^2)	120
F0 contour	7 prosodic statistics	7
Spectral contrast	7 bands × 2 stats	14
Chroma	12 pitch classes × 2 stats	24
Spectral desc.	ZCR, centroid, bandwidth, rolloff, RMS, flatness	6
Total		**411**

3.4 Statistical Validation

Stationarity (ADF test). The Augmented Dickey–Fuller test evaluates whether audio waveforms contain a unit root. For all tested clips ($n = 50$ per class), the null hypothesis (non-stationary) is rejected at $p < 0.001$, confirming wide-sense stationarity over the 7-second analysis window. This justifies frame-based STFT and MFCC extraction.

Gaussianity (Shapiro–Wilk test). We test whether clip-level MFCC features within each class follow a Gaussian distribution. Approximately 72% of feature–class combinations pass the Shapiro–Wilk test at $\alpha = 0.05$. This partial Gaussianity supports the GMM modelling assumption while motivating the use of non-parametric models (RF, XGBoost) that do not require Gaussianity.

Implications for model selection. The stationarity result validates our feature extraction pipeline. The *partial* Gaussianity creates a natural ablation: GMM (assumes Gaussianity) vs. RF/XGBoost (assumption-free), allowing us to quantify how much the Gaussian assumption costs in practice.

3.5 Models

3.5.1 Model 1: Bayesian GMM Classifier

We train one Bayesian Gaussian Mixture Model per class with a Dirichlet process prior on component weights:

$$p(\boldsymbol{\pi}) = \mathrm{Dir}(\alpha_0/K, \ldots, \alpha_0/K) \tag{6}$$

where $\alpha_0 = 1/K$ encourages sparsity. Each class GMM uses up to $K = 8$ components with diagonal covariance. Components with weight < 0.01 are automatically pruned. Classification follows the maximum-likelihood rule:

$$\hat{y} = \arg\max_{c \in \mathcal{C}} \ \log p(\mathbf{x} \mid \boldsymbol{\theta}_c) \tag{7}$$

3.5.2 Model 2: SMOTE-Enhanced SVM

The SVM pipeline addresses class imbalance at three levels:

1. **Data level**: SMOTE generates synthetic minority samples by interpolating between $k = 3$ nearest neighbours in feature space.
2. **Algorithm level**: `class_weight='balanced'` scales the penalty $C_c \propto 1/n_c$.
3. **Decision level**: One-vs-One (OvO) decomposition trains $\binom{5}{2} = 10$ binary classifiers, each on a more balanced sub-problem.

The RBF kernel is:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\bigl(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2\bigr) \tag{8}$$

Hyperparameters $C \in \{1, 10, 100\}$ and $\gamma \in \{\text{scale}, 0.01\}$ are selected via 3-fold cross-validation optimising macro-F1.

3.5.3 Model 3: Left-Right HMM

For the HMM we use frame-level MFCC sequences ($T \times 13$ per clip, using only the first 13 MFCCs to reduce noise). One left-right Gaussian HMM with 8 hidden states is trained per class. The left-right topology enforces:

$$a_{ij} = 0 \quad \text{for } j < i \qquad \text{(no backward transitions)} \tag{9}$$

with self-loop probability 0.7 and forward transition probability 0.3. This mirrors the temporal progression of a cry episode: onset $\rightarrow$ peak $\rightarrow$ offset.

3.5.4 Model 4: Random Forest

500 decision trees with `class_weight='balanced_subsample'`, `max_features='sqrt'`, and `min_samples_leaf=` Feature importance scores $\mathrm{MDI}(f) = \sum_t \Delta\mathrm{Gini}(f, t)$ identify which acoustic features drive classification.

3.5.5 Model 5: XGBoost

300 boosting rounds with max depth 6, learning rate 0.1, and balanced sample weights computed as $w_i = N/(K \cdot n_{y_i})$. Column and row subsampling (0.8 each) provides regularisation.

3.5.6 Model 6: Stacking Ensemble

A meta-learner (Logistic Regression with balanced class weights) combines the probability outputs of SVM, RF, and XGBoost:

$$\mathbf{z}_i = [\hat{p}_{\mathrm{SVM}}(\mathbf{x}_i),\ \hat{p}_{\mathrm{RF}}(\mathbf{x}_i),\ \hat{p}_{\mathrm{XGB}}(\mathbf{x}_i)] \in \mathbb{R}^{15} \tag{10}$$

The meta-learner is trained on the same training set (no cross-validated meta-features for computational efficiency in Phase 1; cross-validated stacking will be used in Phase 3).

3.6 Evaluation Protocol

The dataset is split 80/20 (train/test) with stratification by class and fixed random seed (42) for reproducibility. We report:

- Per-class precision, recall, and F1-score
- **Macro-averaged F1** (primary metric—treats all classes equally regardless of support)
- Weighted-averaged F1
- Overall accuracy
- Matthews Correlation Coefficient (MCC)
- Cohen's Kappa (κ)
- AUC-ROC (one-vs-rest, macro-averaged)
- Normalised confusion matrices

Why macro-F1 over accuracy. With imbalanced classes, accuracy is misleading: predicting "hunger" for every input yields ∼27% accuracy on the augmented set (or ∼84% on the original). Macro-F1 weights each class equally and penalises models that ignore minority classes.

4 Results

4.1 Data Leakage Analysis

Our initial pipeline augmented audio to disk *before* splitting, then applied a random 80/20 train/test split on all files (originals + augmented). This caused augmented variants of the same original recording to appear in both train and test sets. For example, the 8 original burping clips were augmented to 144 files; a random split placed ∼115 in training and ∼29 in testing—but each test sample was a pitch-shifted or time-stretched copy of a training sample.

Table 3: Impact of data leakage: Leaky vs. leak-free evaluation.

Model	**Leaky (Before Fix)**		**Leak-Free (After Fix)**	
	Acc	MacF1	Acc	MacF1
XGBoost	0.965	0.965	0.804	0.178
SVM	0.926	0.921	0.815	0.270
RF	0.929	0.929	0.804	0.179
Ensemble	0.951	0.944	0.837	0.249
GMM	0.622	0.547	0.641	0.207
HMM	0.348	0.310	0.674	0.162

Key observation. Discriminative models (XGBoost, SVM, RF) suffered the largest drops because they had the most capacity to memorise near-duplicate patterns. The leaky XGBoost macro-F1 of 0.965 collapsed to 0.178—a **81.6%** decrease—confirming that the original results were almost entirely driven by test-set contamination. Generative models (GMM, HMM) were less affected because they model class distributions rather than decision boundaries, making them less susceptible to memorising augmented variants.

4.2 Leak-Free Model Comparison

All results below use the corrected split-before-augment protocol. The test set consists of 92 original recordings with no augmented variants.

Table 4: Full evaluation metrics (leak-free, 92-sample test set).

Model	**Acc**	**MacF1**	**WgtF1**	**MCC**	κ	**AUC**
Bayesian GMM	0.641	0.207	0.670	−0.045	−0.044	0.454
SVM (SMOTE)	**0.815**	**0.270**	**0.783**	**0.216**	**0.203**	**0.707**
HMM (left-right)	0.674	0.162	0.678	−0.080	−0.079	—
Random Forest	0.804	0.179	0.751	0.013	0.010	0.632
XGBoost	0.804	0.178	0.746	−0.045	−0.032	0.661
Ensemble	0.837	0.249	0.780	0.155	0.096	0.599

Interpretation. SVM achieves the highest macro-F1 (0.270), while Ensemble achieves the highest accuracy (83.7%). However, the near-zero MCC and κ values for most models indicate that performance is dominated by majority-class (hunger) prediction. The accuracy metric is misleading: a trivial "always predict hunger" classifier would achieve $77/92 = 83.7\%$ accuracy.

4.3 Per-Class Analysis

Table 5: Per-class F1 scores (leak-free evaluation).

Model	**Hunger** (77)	**Belly Pain** (3)	**Burping** (2)	**Discomfort** (5)	**Tiredness** (5)
Bayesian GMM	0.78	0.00	0.00	0.25	0.00
SVM (SMOTE)	0.91	0.00	0.00	0.44	0.00
HMM	0.81	0.00	0.00	0.00	0.00
Random Forest	0.90	0.00	0.00	0.00	0.00
XGBoost	0.89	0.00	0.00	0.00	0.00
Ensemble	0.91	0.00	0.00	0.33	0.00

Critical finding. All models achieve zero F1 on belly_pain (3 test samples), burping (2 test samples), and tiredness (5 test samples). Only SVM recognises discomfort ($F_1 = 0.44$). This reveals the fundamental limitation: with only 2–5 test samples per minority class, classical ML models trained on hand-crafted features lack the representational capacity to generalise from such limited original data, even with training-set augmentation.

4.4 Failure Analysis

Majority-class collapse. All six models exhibit majority-class collapse: they predict "hunger" for nearly all inputs because hunger dominates both training (305/1505 original + augmented) and testing ($77/92 = 83.7\%$). The high accuracy ($\sim$80–84%) masks the fact that minority classes are completely ignored (macro-F1 ≤ 0.27).

Augmentation cannot replace real data. Despite augmenting burping from 8 to 300 training samples, all models achieve $F_1 = 0.00$ on the 2 original burping test samples. The 17 augmentation transforms (pitch shift, time stretch, noise) modify surface-level acoustics but preserve the underlying spectral template. With only 8 original templates, the augmented training set lacks the spectral diversity needed for generalisation.

GMM struggles with high dimensionality. The Bayesian GMM achieves only 64.1% accuracy, with the Dirichlet process collapsing most components to a single effective component per class. The 411-dimensional feature space is too sparse for reliable density estimation with limited data.

SVM is the only model recognising minority classes. SVM achieves $F_1 = 0.44$ on discomfort (the only non-zero minority F1 across all models), likely because the SMOTE + OvO + balanced class-weight triple strategy provides the most robust imbalance handling. The OvO decomposition is particularly beneficial, reducing each sub-problem to a binary decision where the minority class is not overwhelmed.

5 Feature Importance Analysis

Random Forest's MDI (Mean Decrease in Impurity) identifies the most discriminative features:

Table 6: Top 10 most important features (Random Forest MDI, leak-free).

Rank	Feature	Importance
1	spectral_contrast_1_std	0.0127
2	mfcc_2_std	0.0117
3	spectral_contrast_2_mean	0.0108
4	mfcc_9_mean	0.0103
5	voiced_fraction	0.0090
6	cqcc_4_mean	0.0085
7	delta_2_std	0.0079
8	f0_min	0.0078
9	rms_energy	0.0073
10	delta2_7_std	0.0073

Key observations. Spectral contrast features rank highest, suggesting that harmonic structure (distinguishing modal phonation from dysphonation) is the most discriminative acoustic property. The MFCC variability features (std) outrank the mean values, indicating that *temporal instability* of the spectral envelope is more informative than its average shape. The voiced fraction (F_0-based) ranks 5th, confirming the diagnostic value of prosodic features. Notably, feature importances are more uniformly distributed compared to the leaky evaluation, where a few features dominated—consistent with a model that is genuinely learning rather than memorising.

6 Discussion

6.1 Bias–Variance Analysis

The model hierarchy reveals a clear bias–variance tradeoff:

- **GMM**: High bias (Gaussian assumption is only partially satisfied), low variance—performs consistently but with a performance ceiling.
- **SVM**: Moderate bias (kernel flexibility), but sensitive to the C/γ tradeoff and class imbalance.
- **RF/XGBoost**: Low bias (nonparametric), controlled variance via bagging/boosting—best individual performers.
- **Ensemble**: Lowest variance via model averaging, achieving the best generalisation.

6.2 Theoretical Assumptions

I.I.D. assumption and data leakage. Our initial pipeline violated the i.i.d. assumption by allowing augmented variants of the same original to appear in both train and test sets. The corrected split-before-augment protocol ensures strict separation: augmentation is applied only to the training partition, and the test set contains exclusively original recordings. This produces a pessimistic but honest evaluation.

Gaussianity violations. The 28% of feature–class combinations that fail the Shapiro–Wilk test correspond primarily to spectral contrast and chroma features, which exhibit heavy-tailed distributions. This partially explains GMM's lower performance compared to distribution-free methods (RF, XGBoost).

6.3 Limitations

1. **Extreme class imbalance**: The 47:1 ratio between hunger (382) and burping (8) means the leak-free test set contains only 2–5 samples per minority class. Per-class metrics have very high variance at these sample sizes.
2. **Augmentation cannot substitute real data**: All 17 transforms operate on the same spectral templates, limiting diversity. Cross-corpus data (Baby Chillanto, CryCeleb) would provide genuinely new acoustic patterns.
3. **Clip-level features**: Summarising temporal dynamics into per-clip statistics loses information about cry progression patterns that sequence models (LSTM, Transformer) can capture.
4. **Dataset homogeneity**: All clips derive from the Donate-a-Cry corpus. Real-world deployment will face domain shift from background noise, microphone variability, and infant individuality.

6.4 Phase 2 and Phase 3 Roadmap

Phase 2 (Deep Learning). The leak-free results demonstrate that hand-crafted features with classical ML are insufficient for minority-class recognition in this data regime. Phase 2 will train CNN architectures on log-Mel spectrograms (64 mel bands $\times$ 350 time frames), leveraging transfer learning from pretrained audio models (e.g., MobileNetV2, YAMNet) to overcome the limited training data. SpecAugment (time and frequency masking) and mixup augmentation will provide additional regularisation. Focal loss will replace cross-entropy to focus learning on hard minority-class examples.

Phase 3 (Hybrid + Edge Deployment). Decision-level fusion of Phase 1 probabilistic scores with Phase 2 neural embeddings, deployed on ESP32 via TFLite Micro and in browsers via ONNX.js. The hybrid architecture enables an ablation study: Model A (classical ML only, macro-F1 = 0.27) vs. Model B (DL only) vs. Model C (hybrid fusion, target: outperform both).

7 Conclusion

This phase establishes a rigorous, leak-free classical ML baseline for five-class infant cry classification. The key contributions are:

1. A **411-dimensional neuroscience-informed feature vector** combining MFCCs, CQCCs, F0 dynamics, spectral contrast, and chroma—grounded in the acoustic science of infant cry phonation modes.
2. **Identification and correction of data leakage** in the augmentation pipeline, which inflated prior results from macro-F1 = 0.965 to a corrected 0.270—an 81.6% decrease that reveals the true difficulty of this task under extreme class imbalance.
3. **Split-before-augment protocol** ensuring test-set integrity: originals split first, augmentation applied only to training data in memory.
4. **Honest baseline metrics**: SVM (macro-F1 = 0.270, AUC = 0.707) provides the reference point for Phase 2 deep learning improvement.
5. **Six classical models** with comprehensive imbalance handling (SMOTE, OvO, balanced class weights, Bayesian priors), demonstrating that classical ML alone cannot overcome the fundamental data scarcity (8 burping originals).

The data leakage discovery is itself a methodological contribution: it demonstrates the critical importance of split-before-augment protocols in audio classification, particularly when augmentation preserves the underlying spectral template. Phase 2 will introduce CNN architectures on mel-spectrogram representations with transfer learning, and Phase 3 will fuse classical and deep approaches into a hybrid system deployable on edge devices.

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